cyclepoint – hormone

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Load packages

```
library(dplyr)
library(ggplot2)
library(mgcv)
library(gamm4)
library(ggeffects)
```

Load data

```
allhorm <- read.csv("hormdat.csv")
allhorm$hormone <- as.factor(allhorm$hormone)

indhorm <- read.csv("indhorm.csv")
indhorm$hormone <- factor(indhorm$hormone, levels = c("E2","FSH","LH","P4"))
```

Analysis

Classic sample

Includes data collected in several published, open-access datasets (Hidalgo-Lopez et al., 2021; Pletzer & Noachtar, 2023; Taylor et al., 2020)

Run model

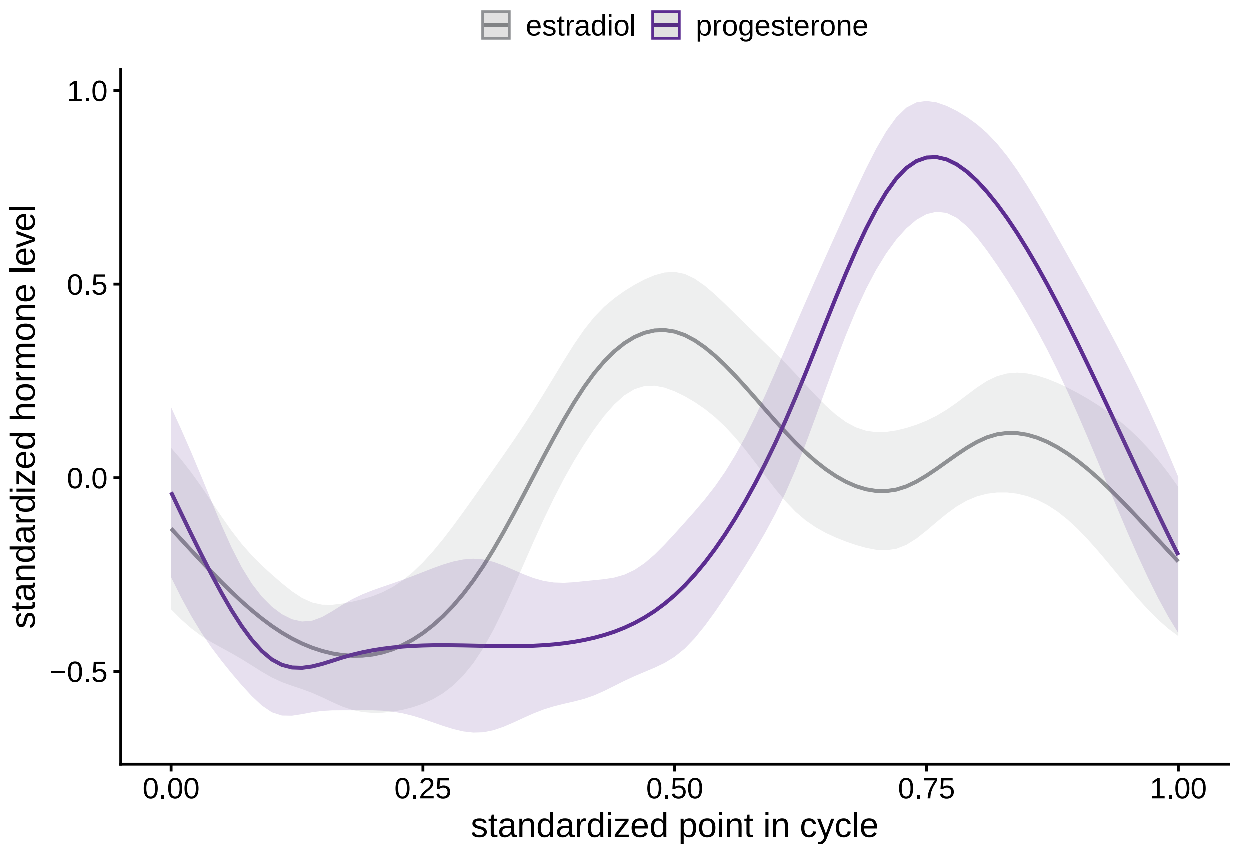
```
cyc <- paste('cyclepoint [',toString(seq(0.0,1.0,by=0.01)),']')

mgh <- gamm4(hormz~s(cyclepoint,bs='cc',by=hormone),
             random=~(1|subject),
             data=allhorm)
summary(mgh$gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## hormz ~ s(cyclepoint, bs = "cc", by = hormone)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.05581    0.03517  -1.587    0.113
##
## Approximate significance of smooth terms:
##                                  edf Ref.df     F p-value
## s(cyclepoint):hormoneestradiol    5.070      8 28.42  <2e-16 ***
## s(cyclepoint):hormoneprogesterone 5.415      8 81.62  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.216
## lmer.REML = 2335.7  Scale est. = 0.42798   n = 1043
```

Plot results

```
plot(ggpredict(mgh$gam,terms=c(cyc,'hormone'))) +
  labs(x='standardized point in cycle',y='standardized hormone level')+
  scale_color_manual(values = c("#8f9194", "#5c2d91"),
                     labels = c("estradiol","progesterone")) +
  scale_fill_manual(values = c("#8f9194", "#5c2d91"),
                     labels = c("estradiol","progesterone")) +
  coord_cartesian(xlim=c(0,1))+
  theme_classic()+
  theme(axis.line=element_line(size=0.5),
        axis.text=element_text(size=11),
        axis.title=element_text(size=13),
        plot.title = element_blank(),
        legend.position='top',
        legend.title = element_blank(),
        legend.text=element_text(size=11),
        legend.margin=margin(t=0,unit='cm'),
        legend.spacing.x=unit(1,'mm'),
        legend.key.size=unit(4,'mm'))
```



Deep phenotyping

Only contains data from an open-source dense-sampling dataset of a single individual (Pritschet et al., 2020).

Run model

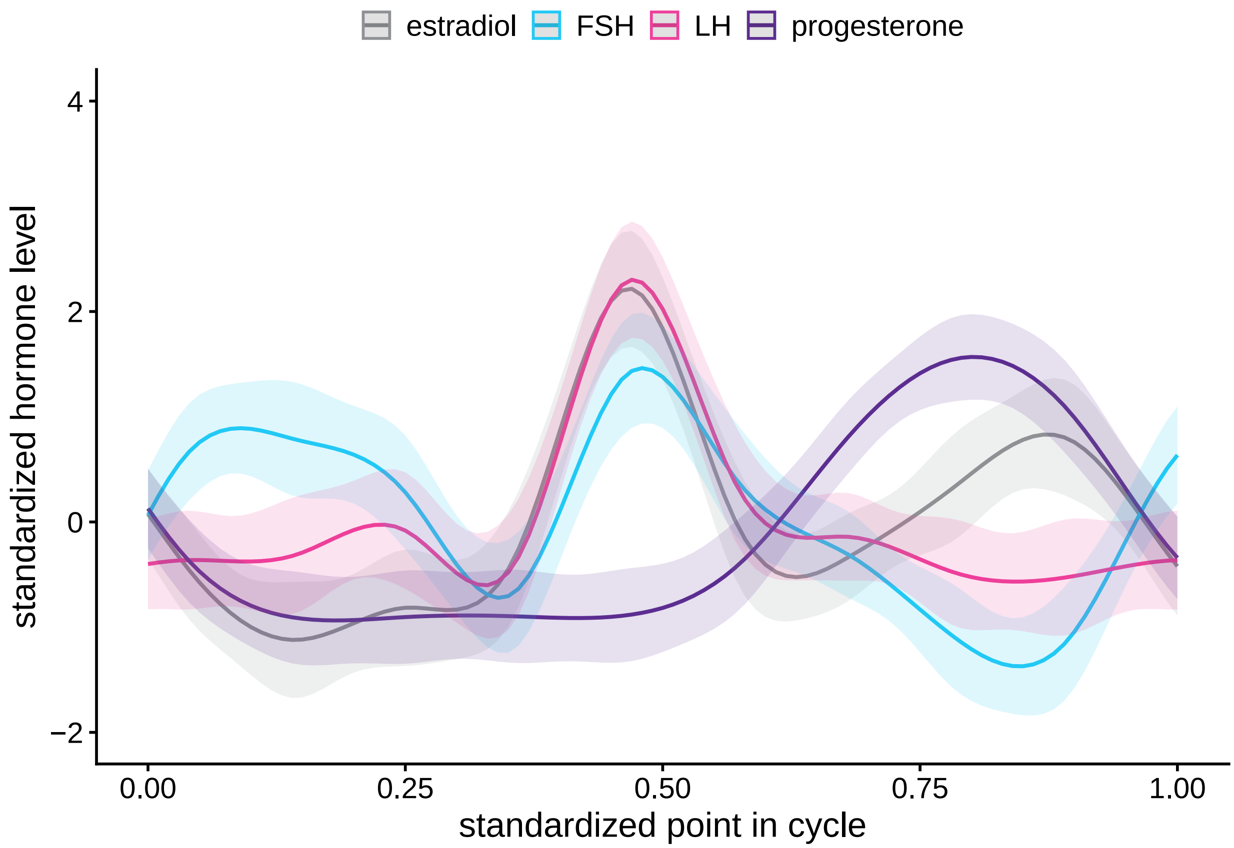
```
mght <- gamm4(hormz~s(cyclepoint,bs='cc',by=hormone),
             data=indhorm)
summary(mght$gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## hormz ~ s(cyclepoint, bs = "cc", by = hormone)
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.849e-12  4.442e-02       0        1
##
## Approximate significance of smooth terms:
##                             edf Ref.df     F p-value
```

```
## s(cyclepoint):hormoneE2  7.037      8 49.20  <2e-16 ***
## s(cyclepoint):hormoneFSH 6.814      8 40.95  <2e-16 ***
## s(cyclepoint):hormoneLH  7.054      8 37.30  <2e-16 ***
## s(cyclepoint):hormoneP4  4.867      8 57.98  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.757
## lmer.REML = 262.75  Scale est. = 0.23678   n = 120
```

Plot results

```r
plot(ggpredict(mght$gam,terms=c(cyc,'hormone'))) +
  labs(x='standardized point in cycle',y='standardized hormone level')+
  coord_cartesian(xlim=c(0,1),ylim=c(-2,4))+
  scale_color_manual(values = c("#8f9194", "#22c9f5", "#ee3f9a", "#5c2d91"),
                     labels = c("estradiol","FSH","LH","progesterone")) +
  scale_fill_manual(values = c("#8f9194", "#22c9f5", "#ee3f9a", "#5c2d91"),
                     labels = c("estradiol","FSH","LH","progesterone")) +
  theme_classic()+
  theme(axis.line=element_line(size=0.5),
        axis.text=element_text(size=11),
        axis.title=element_text(size=13),
        plot.title = element_blank(),
        legend.position='top',
        legend.title = element_blank(),
        legend.text=element_text(size=11),
        legend.margin=margin(t=0,unit='cm'),
        legend.spacing.x=unit(1,'mm'),
        legend.key.size=unit(4,'mm'))
```


estradiol
FSH
LH
progesterone
4
2
0
−2
standardized hormone level
0.00
0.25
0.50
0.75
1.00
standardized point in cycle